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Decomposing MLP Activations into Interpretable Features via Semi-Nonnegative Matrix Factorization

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Abstract

A central goal for mechanistic interpretability has been to identify the right units of analysis in large language models (LLMs) that causally explain their outputs. While early work focused on individual neurons, evidence that neurons often encode multiple concepts has motivated a shift toward analyzing directions in activation space. A key question is how to find directions that capture interpretable features in an unsupervised manner. Current methods rely on dictionary learning with sparse autoencoders (SAEs), commonly trained over residual stream activations to learn directions from scratch. However, SAEs often struggle in causal evaluations and lack intrinsic interpretability, as their learning is not explicitly tied to the computations of the model. Here, we tackle these limitations by directly decomposing MLP activations with semi-nonnegative matrix factorization (SNMF), such that the learned features are (a) sparse linear combinations of co-activated neurons, and (b) mapped to their activating inputs, making them directly interpretable. Experiments on Llama 3.1, Gemma 2 and GPT-2 show that SNMF derived features outperform SAEs and a strong supervised baseline (difference-in-means) on causal steering, while aligning with human-interpretable concepts. Further analysis reveals that specific neuron combinations are reused across semantically-related features, exposing a hierarchical structure in the MLP's activation space. Together, these results position SNMF as a simple and effective tool for identifying interpretable features and dissecting concept representations in LLMs.

1 Introduction

A central question for mechanistic interpretability is how to decompose hidden representations of large language models (LLMs) into constituent parts (Mueller et al., 2024; Geiger et al., 2024; Sharkey et al., 2025). A natural widely-studied unit is individual neural activations. However, while

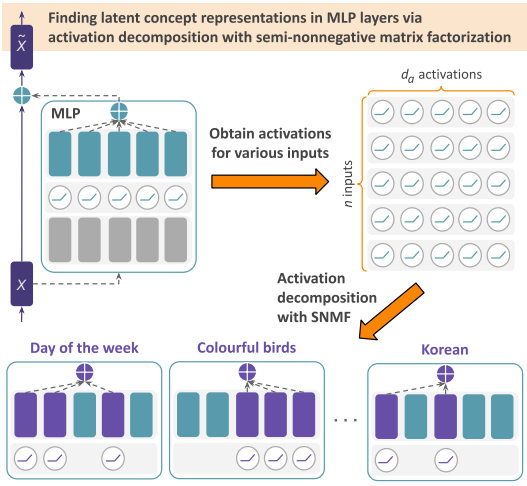

Figure 1: We find latent concept representations by decomposing MLP activations with semi-nonnegative matrix factorization (SNMF). Our method finds sparse compositional MLP features that align with interpretable concepts and outperform existing unsupervised methods and a strong supervised baseline at causal steering.

individual activations often demonstrate human-interpretable concepts (Karpathy et al., 2015; Geva et al., 2021, 2022; Choi et al., 2024), it has long been established that they participate in the representation of multiple intelligible concepts (Smolensky, 1986; Rumelhart et al., 1986; Olah et al., 2020; Bolukbasi et al., 2021; Gurnee et al., 2023).

An alternative unit of analysis that has recently gained consensus is directions in activation space (Elhage et al., 2021; Gurnee et al., 2023; Nanda et al., 2023; Park et al., 2024). A key question is how to find directions encoding interpretable features in an unsupervised manner, namely, without any prior about the concepts they represent. A large body of work has explored dictionary learning methods with sparse autoencoders (SAEs; Bricken et al. 2023; He et al. 2024; Lieberum et al. 2024) to learn directions called *features*. Features are then assigned textual labels with automatic pipelines (Hernandez et al., 2021; Bills et al., 2023; Gur-

Arieh et al., 2025). However, while SAEs can extract interpretable features, they often fail to provide a better unit of analysis in causal evaluations (Huang et al. 2024; Chaudhary and Geiger 2024; Wu et al. 2025; Mueller et al. 2025; cf. Marks et al. 2025). Moreover, the learned features are not constrained to the model's representation space (Menon et al., 2025; Paulo and Belrose, 2025) nor grounded in specific model mechanisms.

In this work, we tackle the problem of finding interpretable features in LLMs by *decomposing MLP activations*. Instead of learning directions from scratch—as in widely-used SAEs trained on residual stream representations—our method represents features as *sparse linear combinations of existing directions in MLP weight matrices*, defined by groups of neurons (see Figure 1). To decompose the MLP hidden activations, we propose using *semi-nonnegative matrix factorization* (SNMF; Ding et al. 2010). We collect MLP activations with d_a dimensions from a large n-token text, and decompose the matrix $A \in \mathbb{R}^{d_a \times n}$ of data into two matrices: a matrix $Z \in \mathbb{R}^{d_a \times k}$ of MLP features that define sparse neuron combinations, and a nonnegative coefficient matrix $Y \in \mathbb{R}^{k \times n}_{\geq 0}$. A fundamental advantage of this approach over SAEs is that the coefficient matrix Y encodes which tokens contributed to the creation of a feature.

We use texts that strongly activate features to assign textual *concept labels*, and evaluate the features along two axes (Huang et al., 2023; Wu et al., 2025): (1) *concept detection*, measuring the ability of the feature to separate texts exemplifying the concept label and neutral texts, and (2) *concept steering*, evaluating how well feature activation steers generated texts toward the concept label while preserving fluency. We find that features learned with SNMF outperform widely used SAEs (Lieberum et al., 2024; He et al., 2024) and supervised features computed with difference-in-means (Marks and Tegmark, 2024; Rimsky et al., 2024; Turner et al., 2024) on concept steering, while performing comparably on concept detection.

Further qualitative analysis shows that MLP features are compositional: recursively applying SNMF to the learned MLP features further composes features of fine-grained concepts (e.g., *Javascript* and *Python*) into features corresponding to more high-level concepts (e.g., *programming*). This showcases the inverse of the feature splitting phenomenon observed in SAEs, where larger SAEs have clusters of features corresponding to individual features of smaller SAEs (Bricken et al., 2023; Chanin et al., 2024). Moreover, semantically-related features (e.g., the different weekdays) share a core set of neurons and are differentiated by exclusive neurons that control the promotion and suppression of fine-grained concepts.

To conclude, we propose decomposing MLP activations with SNMF for finding interpretable features in LLMs. Our experiments show the effectiveness of this approach compared to existing unsupervised methods and a strong supervised baseline, and our analysis provides insights into how neuron compositionality constructs features in MLP layers. We release our code at https://github.com/ordavid-s/snmf-mlp-decomposition.

2 Related Work

Analyzing Neurons in Transformers Geva et al. (2021, 2022) view the MLP output as a collection of updates to the residual stream, where updates correspond to single neurons which often encode human-interpretable concepts. Alternatively, Meng et al. (2022) considers an MLP layer as an associative memory, where all the neural activations form a key that induces the layer's output. Recently, Cao et al. (2025) used clustering to identify meaningful groups of neurons in CNNs. Dalvi et al. (2018); Gurnee et al. (2023) used supervised methods to isolate groups of neurons in transformers that are responsible for promoting specific concepts or tasks. Unlike prior work, we propose an unsupervised approach that identifies features via patterns in MLP activations. We view neuron combinations as causal mediators reused across inputs to represent similar concepts, extending prior work by showing that such reuse is a natural mechanism the model employs to represent concepts.

Concept Representations in LLMs Recent work (Marks and Tegmark, 2024; Park et al., 2024; Nanda et al., 2023) proposes the linear representation hypothesis; concept representations in language models are approximately linear, aligned with specific directions in the residual stream. However, several works challenge this view (Csordás et al., 2024; Park et al., 2025; Engels et al., 2025; Levy and Geva, 2025), proposing that some concepts have meaning when viewed in multiple dimensions. These works demonstrate the presence of structured, complex concept representations. However, they do not explain how such representations are formed or grounded in the model's under-

lying mechanisms. Our work applies SNMF to investigate how neurons form complex features, suggesting a complementary view that defines multi-dimensional concepts through their formation.

Finding Latent Concepts Supervised methods have been shown effective in finding directions in activation space that localize (Geiger et al., 2023; Wu et al., 2023) or steer for (Wu et al., 2024; Marks and Tegmark, 2024) predefined concepts. However, fewer works propose unsupervised methods for finding meaningful directions. Early work applied methods such as clustering (Sajjad et al., 2022) and nonnegative matrix factorization (Yun et al., 2021) to the residual stream. These works provided insight into interpretable concepts utilized by the model, but without finding concrete directions in activation space. More recently, SAEs have been used to learn directions that align with interpretable concepts (Bricken et al., 2023; Cunningham et al., 2023). Unlike SAEs, SNMF offers intrinsic interpretability by linearly decomposing activations into sparse features and a nonnegative coefficient matrix that directly attributes each activation to the features it expresses.

NMF and Deep Learning NMF has previously been used to identify interpretable concepts in deep neural networks. Prior work (Lee and Seung, 1999; Melas-Kyriazi et al., 2022; Oldfield et al., 2023) has demonstrated NMF isolates meaningful parts in images, e.g., eyes and noses, by factorizing image data. Yun et al. (2021) applied NMF to activations in the residual stream, showing they encode interpretable concepts. Here, we extend this approach to MLP activations and show that the discovered MLP features have causal roles within the model.

3 Decomposing MLP Activations with Semi-Nonnegative Matrix Factorization

We propose an unsupervised method for finding meaningful directions in LLMs, focusing on their MLP layers. Our approach is motivated by the view that the MLP output to the residual stream can be expressed as a linear combination of underlying features, each formed by a specific group of neurons. Their activations specify how they are linearly combined to construct concepts in a given input. Thus, inputs that express similar concepts should activate similar neurons. This theory is further supported by Yun et al. (2021) who provide evidence of the additive nature of concepts in the model's residual stream.

Following this view, we propose to identify groups of neurons representing interpretable features using semi-nonnegative matrix factorization (Ding et al., 2010). SNMF factorizes a data matrix into two matrices: a matrix of *features* and a nonnegative *coefficient matrix* that maps between inputs and features. Our choice of SNMF is motivated by Lee and Seung (1999) who demonstrated that the nonnegativity constraints in NMF encourage parts-based, additive representations, which tend to yield interpretable features. SNMF retains this interpretability while relaxing the constraint for nonnegativity in the features, thus being a better fit for MLP activations.

We provide notation and explain how we apply SNMF to find interpretable features in MLP layers.

MLP Activations We assume a transformer-based language model with L layers, a hidden dimension d, and an MLP inner-dimension d_a. An MLP layer is defined with two parameter matrices $W_K, W_V^T \in \mathbb{R}^{d_a \times d}$ and an element-wise nonlinear activation function σ:[1]

$$\texttt{MLP}(\mathbf{h}) = W_V \sigma(W_K \mathbf{h}) := W_V \mathbf{a} \tag{1}$$

where $\mathbf{h} \in \mathbb{R}^d$ is an input hidden representation and $\mathbf{a} \in \mathbb{R}^{d_a}$ is a vector of activations.[2] Let $\mathbf{k}_i$ be the i-th row of W_K and $\mathbf{v}_i$ the i-th column of W_V, the output of the MLP layer can be cast as a linear combination of W_V's columns, each weighted by its corresponding activation:

$$\texttt{MLP}(\mathbf{h}) = \sum_{i=1}^{d_a} \sigma(\mathbf{k}_i \cdot \mathbf{h})\mathbf{v}_i = \sum_{i=1}^{d_a} a_i \mathbf{v}_i \tag{2}$$

Decomposing Activations with SNMF Given neuron activations $A := [\mathbf{a}_1 \dots \mathbf{a}_n] \in \mathbb{R}^{d_a \times n}$ obtained from an MLP layer for a set of n corresponding input representations $\mathbf{h}_1, ..., \mathbf{h}_n$, we apply SNMF to decompose A into MLP features. Specifically, A is factorized into a product of two matrices $Z \in \mathbb{R}^{d_a \times k}$ and $Y \in \mathbb{R}_{\geq 0}^{k \times n}$ such that:

$$A \approx ZY, \tag{3}$$

where k is a hyperparameter defining the number of features.

The columns of Z define the representations of the *MLP features*. Each MLP feature $\mathbf{z}_i \in \mathbb{R}^{d_a}$ lies

[1] Bias terms are omitted for brevity.

[2] In modern LLMs, activations often go through additional gating before the output projection (Liu et al., 2021).